



Can Cross-Layer Transcoders *replace* Vision Transformer activations? *An interpretable perspective on vision.*

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TL;DR

Cross-Layer Transcoders reconstruct and replace ViT MLP activations while preserving zero-shot accuracy, enabling faithful attribution of which layers build the final [CLS] and patch-token representations

Motivation

WHY INTERPRETABILITY

Vision Transformers achieve strong zero-shot performance, but their **internal representations** remain **difficult to interpret** [4]

MECHANISTIC INTERPRETABILITY

SAEs and Transcoders operate **within a single layer**, interpreting features **locally** but not explaining how information is transformed **across depth** [2,3]

IDEA: CLTs FOR VISION

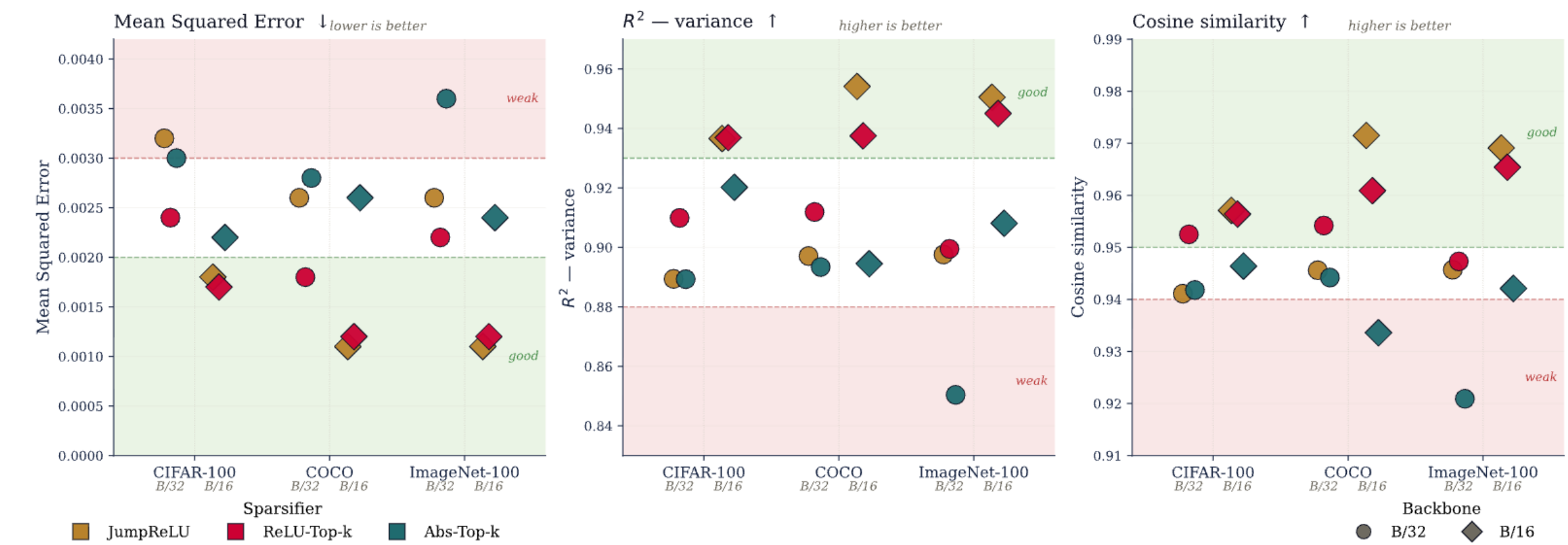
Leveraging the notion of CLTs from language to vision [1]: ViTs combine **spatial patch** tokens, **2D structure**, variable **granularity**, and a **global [CLS]** representation

Can CLTs faithfully replace ViT computations and reveal how visual representations are built across ViT depth?

RQ 01 · FUNCTIONAL REPLACEMENT

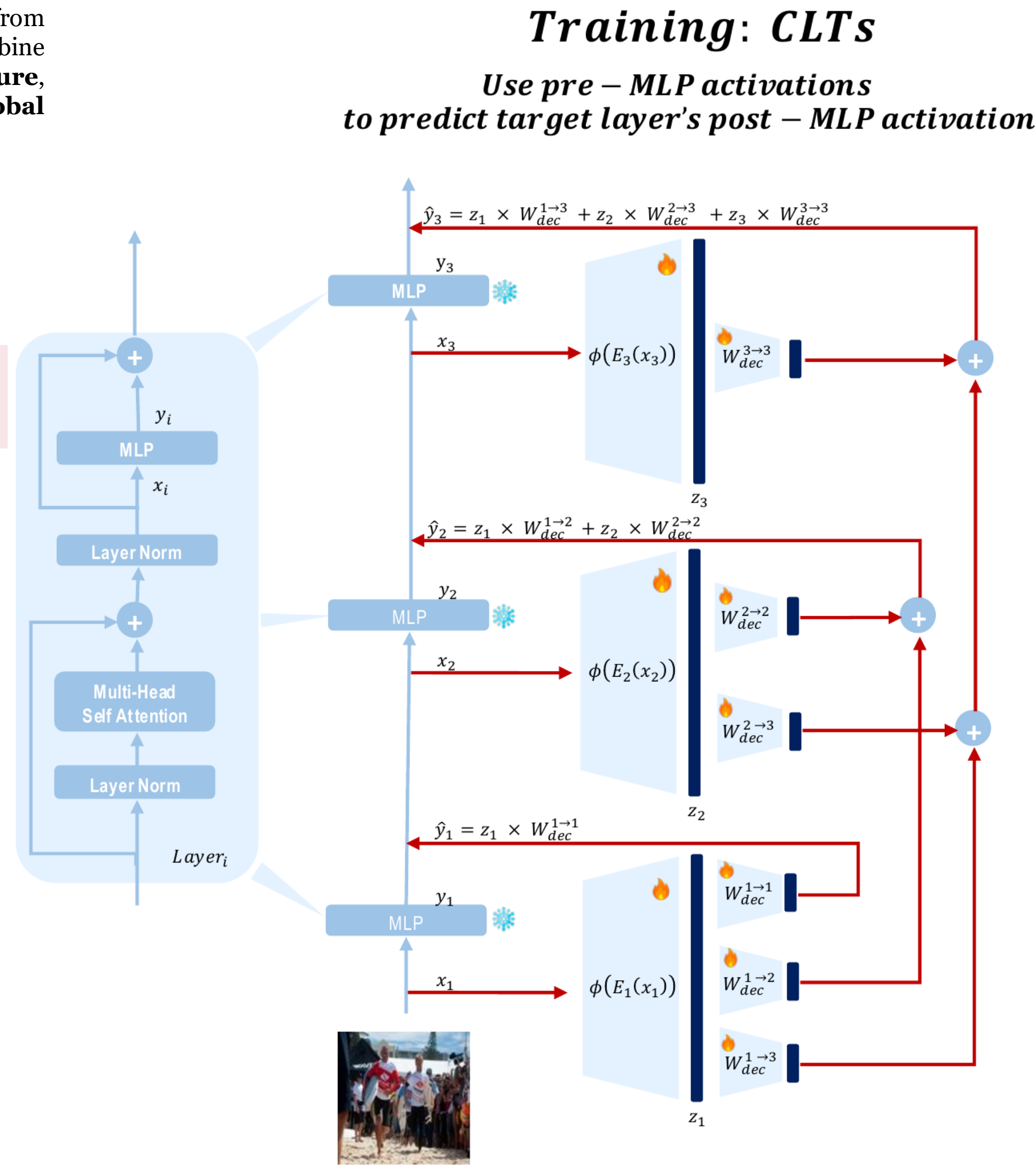
Can CLTs functionally replace MLP blocks in ViTs as an alternative interpretable proxy?

Reconstruction of MLP Activations



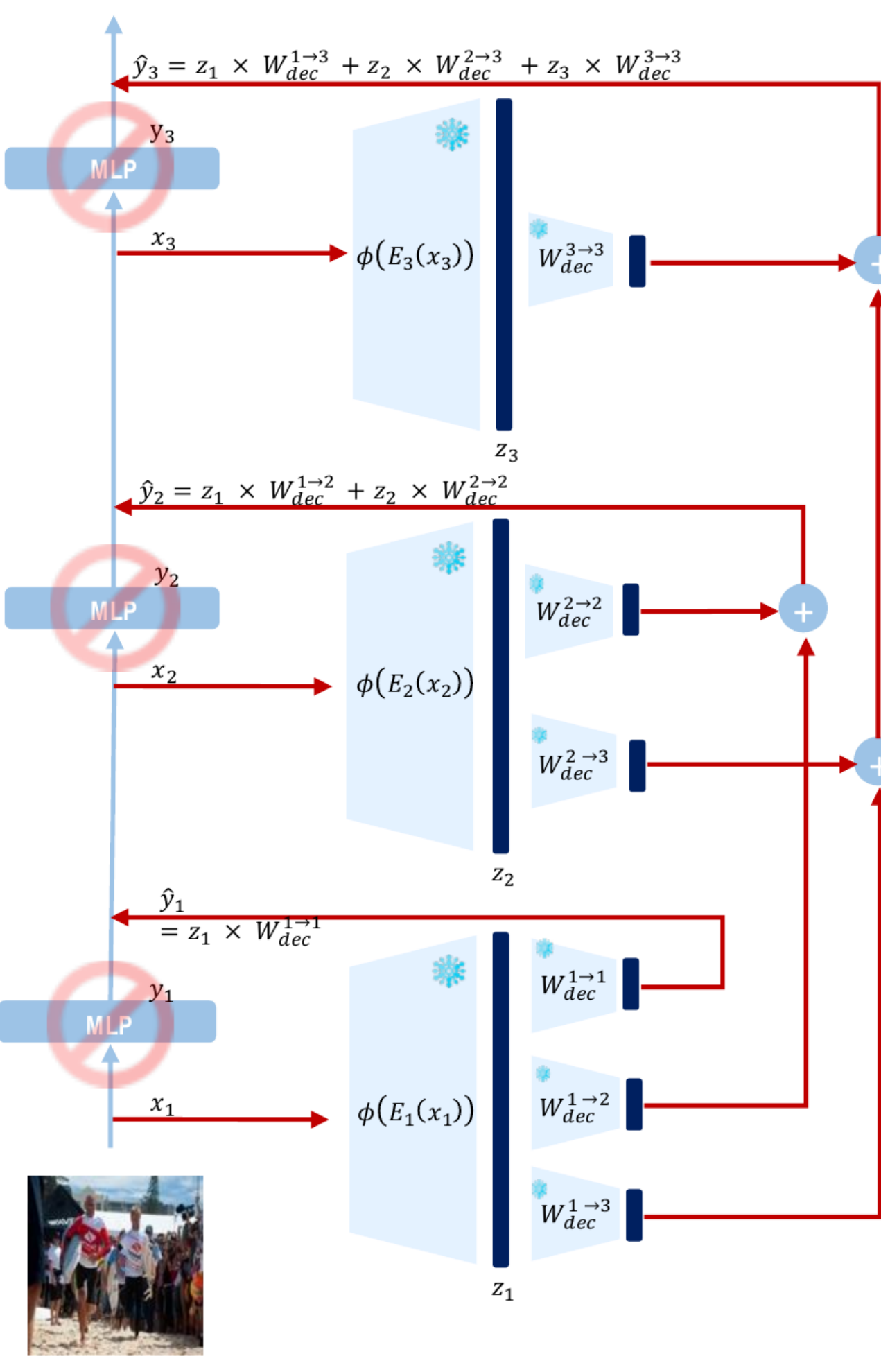
Insight 01 · FUNCTIONAL REPLACEMENT based on Token Size

Patch granularity matters. ViT-B/16 reconstruction more faithful than ViT-B/32. Smaller patches distribute information across more tokens, yielding simpler per-token activations that are easier to approximate



Inference: Replacement Model

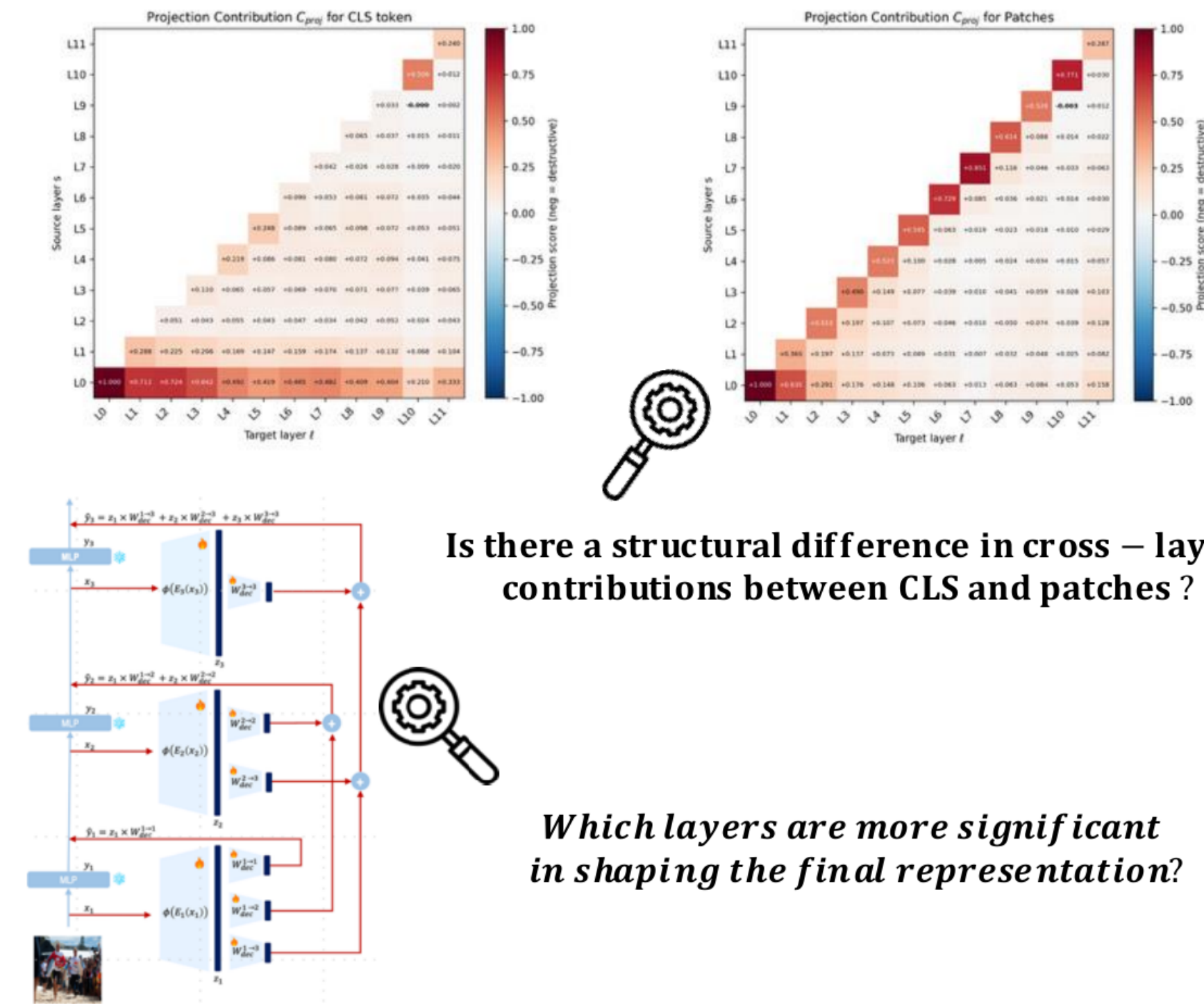
Replace MLP blocks with learned CLTs



ViT Understanding through CLTs

Use CLTs as an interpretable proxy for ViTs

$$C_{i \rightarrow j}^{proj} = E_x \left[\left(\frac{1}{T} \sum_{t=1}^T \frac{\langle c_{i \rightarrow j}^{(t)}, \hat{y}_j^{(t)}(x) \rangle}{\|\hat{y}_j^{(t)}(x)\|_2} \right)^2 \right]$$



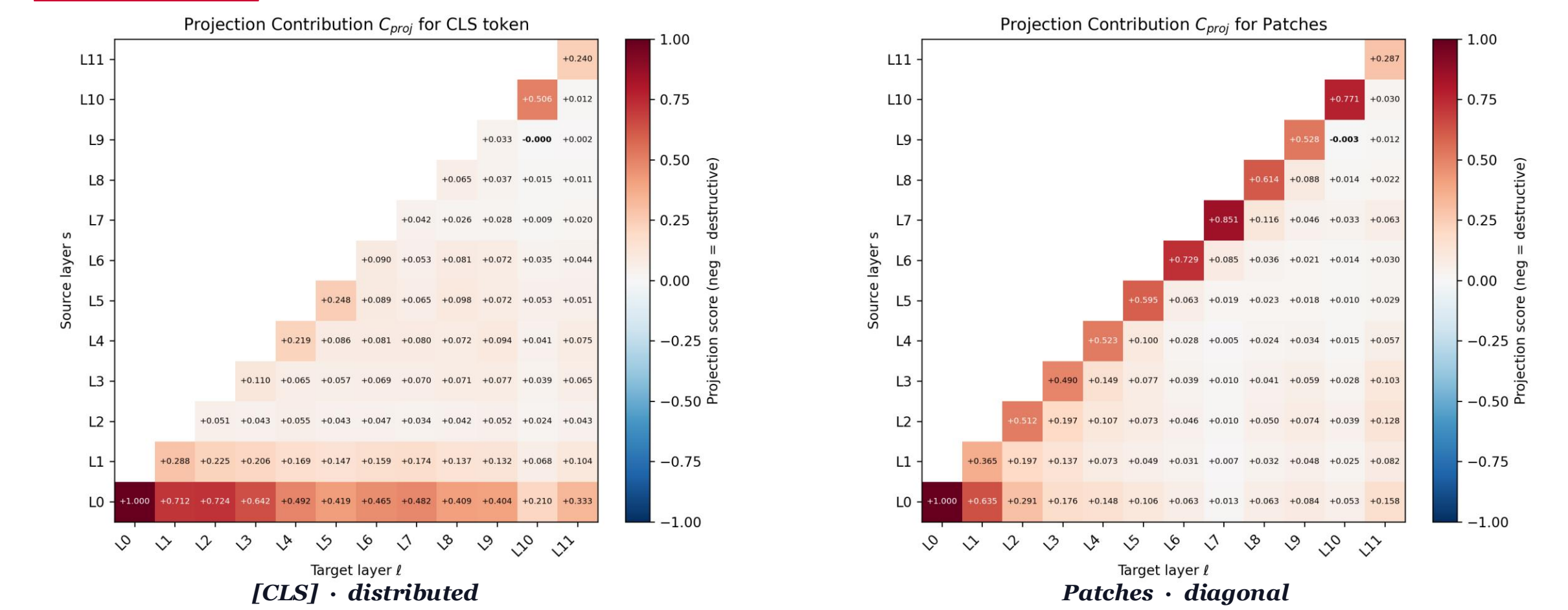
Is there a structural difference in cross-layer contributions between CLS and patches?

Which layers are more significant in shaping the final representation?

RQ 02 · FAITHFUL ATTRIBUTION

Do cross-layer contribution scores yield faithful, process-level attribution?

CROSS-LAYER CONTRIBUTION SCORES



Insight 03 · CROSS-LAYER CONTRIBUTION SCORES based on Token Type

[CLS] integrates across depth; patches remain local. Patch tokens show diagonal attribution, while [CLS] draws credit from many preceding layers

FAITHFULNESS ATTRIBUTION

Dataset	Tokens	Baseline	NECESSITY remove top-1 attributed Layer		SUFFICIENCY retain only top-4 Layers	
			Drop Top-1	Keep Top-4	Drop Top-1	Keep Top-4
CIFAR-100	all	61.65	59.68	-1.97%	59.96	-1.69%
	[CLS]	61.65	59.91	-1.74%	60.72	-0.93%
COCO	all	43.12	42.98	-0.14%	43.00	-0.10%
	[CLS]	43.12	42.78	-0.34%	43.00	-0.12%
ImageNet-100	all	80.42	74.94	-5.48%	80.56	+0.14%
	[CLS]	80.42	74.86	-5.56%	80.48	+0.06%

Insight 04 · NECESSARY AND SUFFICIENT ATTRIBUTION LAYERS

The top-4 attributed layers recover accuracy, while removing the highest-scored layer causes substantial degradation

CONCLUSION

CLTs make ViT computation **interpretable across depth**. They **preserve model behavior** while revealing that patch tokens are mostly layer-local, whereas [CLS] integrates information broadly across layers.

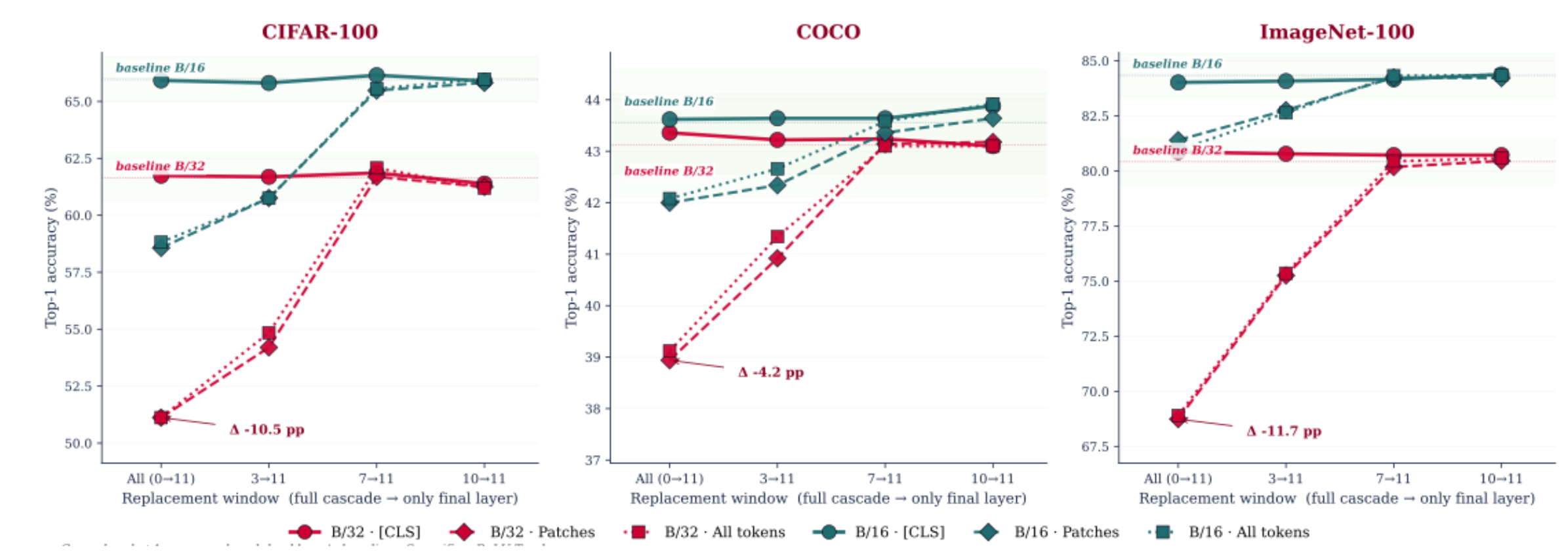
FUTURE WORK

From interpretation to intervention: extend CLTs to attention and larger VLMs, using sparse cross-layer features for circuit discovery and adversarial robustness.

REFERENCES

- [1] Ameisen et al., Circuit tracing, 2025.
- [2] Duněský et al., Transcoders find interpretable LLM feature circuits, 2024.
- [3] Bricken et al., Towards monosemanticity, 2023.
- [4] Radford et al., Learning transferable visual models from natural language supervision, 2021.

Cascaded Replacement



Insight 02 · FUNCTIONAL REPLACEMENT based on Token Type

CLTs can replace MLP blocks, especially in later layers or for [CLS] tokens across all layers, preserving and even improving in some cases zero-shot classification performance

CROSS-LAYER FEATURE RETRIEVAL

